

Module 3 session-6

Encapsulation

Encapsulation means **keeping data private and controlling access to it through methods**.

For example, in Instagram, your account information can be treated like private data. You don't directly change the internal data of your account. You use options such as **Edit Profile**, which internally updates the information.

In our C++ example:

private:

```
string phoneNumber;
```

and we access it using:

```
setPhoneNumber()
```

```
getPhoneNumber()
```

Abstraction

Abstraction means **hiding unnecessary internal details and showing only what the user needs**.

For example, when you upload a photo on Instagram, you simply press **Post**. You don't need to know how Instagram stores the image, processes it, or sends it to its servers.

In C++:

```
virtual void upload()=0;
```

The user only knows that `upload()` is available. The actual implementation is provided by classes like Electronics and Clothing.

Simple difference

Encapsulation

Hides/protects data

Achieved using private, protected, getters/setters

Abstraction

Hides implementation details

Achieved using abstract classes and virtual functions

Encapsulation

Example: Private phone number

Focuses on **data security**

Abstraction

Example: upload() method

Focuses on **what the object does**